

MULTI-QUEUE:

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#include<stdio.h>
#define MAX 20
struct Process{int pid,at,bt,ct,tat,wt,type;};
int main(){
    printf("riya gupta , usn:1WA24CS235 \n");
    struct Process p[MAX],sq[MAX],uq[MAX];
    int n,i,sq_count=0,uq_count=0,time=0;
    printf("Enter number of processes: ");
    scanf("%d",&n);
    for(i=0;i<n;i++){
        printf("\nProcess %d\n",i+1);
        p[i].pid=i+1;
        printf("Arrival Time: ");
        scanf("%d",&p[i].at);
        printf("Burst Time: ");
        scanf("%d",&p[i].bt);
        printf("Type (1=System,2=User): ");
        scanf("%d",&p[i].type);
        if(p[i].type==1) sq[sq_count++]=p[i];
        else uq[uq_count++]=p[i];
    }
    for(i=0;i<sq_count;i++){
        if(time<sq[i].at) time=sq[i].at;
        time+=sq[i].bt;
        sq[i].ct=time;
    }
    for(i=0;i<uq_count;i++){
        if(time<uq[i].at) time=uq[i].at;
        time+=uq[i].bt;
        uq[i].ct=time;
    }
    float total_tat=0,total_wt=0;
    printf("\n---System Queue---\n");
    printf("PID\tAT\tBT\tCT\tTAT\tWT\n");
    for(i=0;i<sq_count;i++){
        sq[i].tat=sq[i].ct-sq[i].at;
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sq[i].wt=sq[i].tat-sq[i].bt;
total_tat+=sq[i].tat;
total_wt+=sq[i].wt;
printf("%d\t%d\t%d\t%d\t%d\t%d\n",sq[i].pid,sq[i].at,sq[i].bt,sq[i].ct,sq[i].tat,sq[i].wt);
}
printf("\n---User Queue---\n");
printf("PID\tAT\tBT\tCT\tTAT\tWT\n");
for(i=0;i<uq_count;i++){
uq[i].tat=uq[i].ct-uq[i].at;
uq[i].wt=uq[i].tat-uq[i].bt;
total_tat+=uq[i].tat;
total_wt+=uq[i].wt;
printf("%d\t%d\t%d\t%d\t%d\t%d\n",uq[i].pid,uq[i].at,uq[i].bt,uq[i].ct,uq[i].tat,uq[i].wt);
}
printf("\nAverage Turnaround Time=%.2f",total_tat/n);
printf("\nAverage Waiting Time=%.2f\n",total_wt/n);
return 0;
}

```

Output

riya gupta , usn:1WA24CS235
Enter number of processes: 4

Process 1
Arrival Time: 0
Burst Time: 4
Type (1=System,2=User): 1

Process 2
Arrival Time: 0
Burst Time: 3
Type (1=System,2=User): 1

Process 3
Arrival Time: 0
Burst Time: 8
Type (1=System,2=User): 2

Process 4
Arrival Time: 10
Burst Time: 5
Type (1=System,2=User): 1

---User Queue---

PID	AT	BT	CT	TAT	WT
3	0	8	23	23	15

Average Turnaround Time=9.75

Average Waiting Time=4.75